Customer – Transactions Activity & Spending Patterns

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4th February 2026

About Data

- Customer Demographics ~276K records
- Transactions ~8M records

```
con.execute("DESCRIBE customers").df()
```

	column_name	column_type	null	key	default	extra
0	CustomerID	VARCHAR	YES	None	None	None
1	Individual	VARCHAR	YES	None	None	None
2	Age	DOUBLE	YES	None	None	None
3	Gender	VARCHAR	YES	None	None	None
4	NAICSCode	DOUBLE	YES	None	None	None
5	OriginalCustomerDate	DATE	YES	None	None	None
6	RelationshipYears	DOUBLE	YES	None	None	None
7	RelationshipMonths	DOUBLE	YES	None	None	None
8	BangorWealth	BOOLEAN	YES	None	None	None
9	Payroll	BOOLEAN	YES	None	None	None
10	Merchant	BOOLEAN	YES	None	None	None
11	TPS	BOOLEAN	YES	None	None	None
12	PlingCustomer	BOOLEAN	YES	None	None	None
13	ABLECustomer	BOOLEAN	YES	None	None	None
14	TimeDepositAccount	VARCHAR	YES	None	None	None
15	DepositAccount	VARCHAR	YES	None	None	None
16	LoanAccount	VARCHAR	YES	None	None	None
17	CreditCardAccount	VARCHAR	YES	None	None	None
18	ActiveATMCard	VARCHAR	YES	None	None	None
19	ActiveDebitCard	VARCHAR	YES	None	None	None
20	NumberActiveDDAs	DOUBLE	YES	None	None	None
21	NumberActiveTimeDeposits	DOUBLE	YES	None	None	None
22	NumberActiveLoans	DOUBLE	YES	None	None	None
23	NumberCreditCardAccts	DOUBLE	YES	None	None	None
24	PrimaryBankingCustomerFlag	VARCHAR	YES	None	None	None

```
con.execute("DESCRIBE transactions").df()
```

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	column_name	column_type	null	key	default	extra
0	ATM_Address	VARCHAR	YES	None	None	None
1	ATM_City_State	VARCHAR	YES	None	None	None
2	Merchant_Category	VARCHAR	YES	None	None	None
3	Merchant_Name	VARCHAR	YES	None	None	None
4	Transaction_Code	BIGINT	YES	None	None	None
5	Transaction_Code_Description	VARCHAR	YES	None	None	None
6	Transaction_Type	VARCHAR	YES	None	None	None
7	Transaction_Type_Description	VARCHAR	YES	None	None	None
8	Transaction_Number	VARCHAR	YES	None	None	None
9	Amount_Completed	DOUBLE	YES	None	None	None
10	Transaction_Date	DATE	YES	None	None	None
11	Time_Local_hhmmss	BIGINT	YES	None	None	None
12	Recurring_Trxn	VARCHAR	YES	None	None	None
13	CustomerID	VARCHAR	YES	None	None	None

EDA So Far...

- Customers overview
- Age Distribution (avg - 41.9)
- Relationship with Bank (avg - 13.3 years)
- Various products/holdings (debit card – 65%, credit card – 2%)
- Gender Distribution
- Types of Accounts
- Primary Bank Ratio

- Transactions data range - primarily Nov onwards (10 transactions Sept/Oct)
- Amount distribution (avg - $49.5)
- Weekly/Hourly Classification (maximum on Fridays | general peak time 12 to 1 PM)
- Merchant Categories (Top 3 : 5411 (grocery), 5814 (fast food), 5541 (service stations)
- Transaction types
- Transactions per customers (avg – 64 per customer)

Story/Direction

- Initial –

Bank transactions generally record everything and though they are masked or concise, a bank employee would be able to decode a person's habits and most of their routine based on their bank transactions. Analyzing the different kinds of transactions and deducing patterns in them can also help us in finding ***anomalies based on unusual activity****. This could indicate* ***account takeover or card fraud****. The primary goal is to demonstrate how customer data and their transaction behavior can help us in surfacing this suspicious activity. As more progress is made on this data, the scope is subject to change based on the initial findings.*

- Now –

a customer intelligence system that ***segments customers by their transaction behavior and predicts spending patterns.*** *This would enable more personalized service, targeted product offerings, and deeper understanding of customer needs.*
This can complement the conversation/recommendation engine.

Story/Direction

What this can look like –

- *Customer Profiling and Segmentation*
  Behavioral clustering based on transaction patterns and demographics
- *Spending Predictions:*
  Spending Tier Movement – Will customer move up/down/stay stable?
  Spending Trajectory – Is spending increasing or declining over time?
  Spending Category/Merchants

Need to validate -

- Customer pool
- Duplicate customerID
- Non-individual type customers with gender value (M,F, _, O, U)
- 1.17% - $0 transactions (failures/refunds?)
- Many customers with age above 100 (individuals and non-individuals; max -126)
- ATM_City_State: 928 missing – D/C Settlements?